Counting Sort

Counting Sort is a **non-comparison-based sorting algorithm** used for sorting integers when the range of input values is known and not too large. Instead of comparing elements, it counts the number of occurrences of each value and uses these counts to place elements in their correct sorted positions.

Key Idea: Count how many times each value appears, then reconstruct the sorted array.

Algorithm

Suppose the input array is:

```
A = [4, 2, 2, 8, 3, 3, 1]
```

Step 1: Find the maximum value

Maximum = **8**

Step 2: Create the count array

Create a count array of size **max + 1 = 9**.

Initially:

Value	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Count	0	0	0	0	0	0	0	0	0

Step 3: Count the occurrences

Value	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Count	0	1	2	2	1	0	0	0	1

Step 4: Compute cumulative counts

Value	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**
Count	0	1	3	5	6	6	6	6	7

These cumulative counts tell the final position of each element.

Step 5: Build the output array

Traverse the original array **from right to left** to maintain stability.

Final sorted array:

```
[1, 2, 2, 3, 3, 4, 8]
```